Energy, Trace, and Laurent-Defect Certificates for Removable Boundaries and Circle-Domain Rigidity

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Abstract

We study the specific conductivity, saved-barrier, and length–area mechanisms in the September 23 manuscript on removable boundaries and circle-domain rigidity. We prove a variable-contrast energy identity, a constant-one equilibrated-flux certificate and a two-sided interval for the first Laurent coefficient gain. These provide finite-contrast leakage bounds and a precisely conditional componentwise high-contrast limit. A weighted two-sided slice estimate improves the manuscript's barrier constant by a factor of four and tolerates certified trace errors. Retaining the full Laurent area defect gives a finite-tail rigidity budget controlling disk-radius mismatch, derivative-modulus error, and the normalized map and its derivative on exterior regions. Compatible Ram moments realize the physical quadratic forms; finite probes without complementary bounds do not certify them. Every local claim has explicit analytic hypotheses and is separated from the companion retained-test construction, actual-image identification, continuous Sobolev removability, and all-component limiting arguments. The contemporary public literature still records the general implication as open, so no independent global resolution is asserted here. Contribution assessment: C3/5, the proved source-specific energy and finite-tail stability certificates are notable local quantitative improvements, not a new global rigidity theorem or a claim of historical novelty.

1 Scope, direction, and current status

A circle domain is a connected nonempty open subset of $\widehat{\mathbb{C}}$ whose complementary components are closed round disks or singletons. The complement may be empty and the collection of singletons may be uncountable. A compact set E is conformally removable if every orientation-preserving sphere homeomorphism conformal on $\widehat{\mathbb{C}} \setminus E$ is Möbius. A circle domain is rigid if every conformal equivalence onto another circle domain is the restriction of a Möbius map. In the first definition a global homeomorphism is already available; in the second it is not.

The pinned source [1] asserts

$$\partial\Omega \text{ removable} \quad \Longrightarrow \quad \Omega \text{ rigid}$$

without a connectivity restriction. It imports the scalar retained-test transfer and selection statements from a companion uniformization manuscript, not merely the classical finite-connectivity theorem. Our literal reading covers all its cached build modules, bibliography, figure source, and README. This is not an independent proof audit of the uninspected companion construction or all cited analytic theorems.

Public status checked October 8, 2026. Rajala's published counterexample [4] disproves *rigid implies removable*, not the displayed direction. More importantly, the September 30 survey of Ntalampekos and Younsi [3, Question 6.1] still asks the displayed question, even for Cantor boundaries. Its discussion identifies the possibility of a point end opening into a nondegenerate target component as an obstruction. Thus the cached September 23 claim is not treated here as a settled consequence of the independent public literature. The discrepancy is reported, not resolved by a numerical certificate or by a date comparison. The survey's CNED and quasihyperbolic results have additional hypotheses.

Conventions. All scalar energies below are real. $D_R = \{z : |z| < R\}$, and planar Dirichlet integrals use Euclidean area. The global function norm on $W^{1,2}(\widehat{\mathbb{C}})$ uses spherical area. An exterior normalized conformal map has expansion $f(z) = z + \sum_{n\geq 1} \alpha_n z^{-n}$, with no constant term. No rescaling of the first Laurent coefficient, of π, or of disk radii is implicit. A numerical upper bound is a certificate only when its integration, geometric coverage, and infinite remainder are justified.

2 The source's dependency architecture

The following map gives proof obligations, not independent certifications.

1. *Ends and normalization.* `preliminaries.tex` constructs a correspondence between *every* complementary component using a monotone quotient sphere and nested physical contours. The inverse collapse map is continuous when the source omitted set is totally disconnected. Exterior area bounds prevent collapse of normalized conformal maps and bound the whole omitted set.

2. *Exactly retained scalar models.* `models.tex` assumes the companion's Corollary 7.6 and Theorem 7.5. For a finite compatible list u_i, compatibility is $|(u_i, u_j)(B)|_2 = 0$ for $i \neq j$. On one shared exhaustion the transferred tests agree *exactly* near the retained cores, have vanishing off-core energy and square-summable hole oscillations, and preserve contour sides. The source expressly imports that the limit's *actual image* is a circle domain.

3. *Values on positive-area omitted dust.* Filling holes and weak compactness give a Sobolev limit. Saturation of conformally invariant energy gives zero gradient on the complement, but not its values there. Source-contour excess tests, finite covers at every scale, and continuity of the collapse map identify the prescribed values on all components, including point components of positive total area.

4. *A near-extremal height.* `conductivity.tex` nearly maximizes $\operatorname{Re} b(h)$ over models retaining the fixed list. Principal quasiconformal deformations preserve that model class. Their exact energy identity, coefficient upper bound, coordinate maximum principle, and logarithmic oscillation estimate give a continuous limiting height. The coarea argument then proves pair-nullity on full pointwise level sets.

5. *Saved barriers and Sobolev extension.* `sobolev.tex` flattens countably many scalar values and uses priority interpolation, with no disjointness assumption on its neighborhoods. Subsequent changes preserve strict plateau and energy margins. The limits are handled by $\forall k\, \exists n_0(k)\, \forall n \geq n_0(k)$ for each saved contour. A line payment rules out *every* nonpoint limiting component. Only then does removability identify the normalized map with the identity.

6. *Common point traces.* `traces.tex` masks every disk-containing source–target pair, including point-to-disk and disk-to-point pairs. Ball synthesis, continuity at accumulating components, and the preceding Sobolev theorem supply ACL restrictions. Countable mask systems and full linewise tails yield null projections for all common point traces on almost every relevant line.

7. *Flux and equality.* `rigidity.tex` bounds slice increments, controls a tail-component multiplicity of two, integrates both axial directions, and then lets the complete disk-pair list exhaust. Source and target areas force a sum of nonnegative defects to vanish. The target dust need not have area zero; its area is omitted in the target inequality.

The source's stronger statements remain indispensable. In particular, zero finite residuals do not prove model existence; local uniform map convergence alone does not identify a prescribed limiting image; a finite disk list does not cover an uncountable point set; and conformal removability cannot be applied to a map that has not yet been shown to extend as a sphere homeomorphism.

3 Variable conductivity and coefficient gain

Let $F \subset D_{R_0}$ be compact and let k be measurable with $1 \le k \le K < \infty$ and $k = 1$ off F. Write $e = (0, 1)$ and let $\mathcal{H}$ be the completion of $C_c^\infty(\mathbb{R}^2; \mathbb{R})$ in the norm $\|\nabla\psi\|_2$, modulo constants in a function realization. It is convenient to regard $\mathcal{H}$ as its closed gradient subspace in $L^2(\mathbb{R}^2; \mathbb{R}^2)$. Set

$$a_k(u, \psi) = \int k\nabla u \cdot \nabla\psi, \qquad \|u\|_k^2 = a_k(u, u), \qquad b = (k - 1)e, \qquad B_k = \int (k - 1).$$

Both b and B_k are finite. Coercivity gives a unique gradient $w \in \mathcal{H}$ satisfying

$$a_k(w, \psi) = -\int b \cdot \nabla\psi \quad (\psi \in \mathcal{H}). \tag{3.1}$$

Write $v = Y + w$ when a representative is fixed. This is a whole-plane transmission problem, not a collection of independent hole problems.

Proposition 3.1 (Variable-contrast identity)**.** *Suppose $\phi = a + iv$ is a normalized orientation-preserving plane homeomorphism, locally $W^{1,2}$, satisfying*

$$a_X = kv_Y, \qquad a_Y = -kv_X, \qquad \phi(z) = z + c/z + O(z^{-2})$$

and conformal off F. Assume the area formula for ϕ and the usual smooth exterior Laurent expansion. Then its $w = v - Y$ solves (3.1), belongs to $\mathcal{H}$, and

$$\Phi_k(w) := \int |\nabla w|^2 + \int (k - 1)|e + \nabla w|^2 = B_k - \|w\|_k^2 = 2\pi \operatorname{Re} c \ge 0. \tag{3.2}$$

If $\phi \circ h$ is an admissible retained-test model and $\operatorname{Re} b(h) > \sup_{\mathcal{M}} \operatorname{Re} b - d$, then $\Phi_k(w) \le 2\pi d$.

Proof. Commuting the weak mixed derivatives of a gives $\operatorname{div}(k\nabla v) = 0$, hence (3.1) for compact tests. Exterior decay gives $w = O(|z|^{-1})$, $\nabla w = O(|z|^{-2})$ there. Cut off at radius R, then mollify; the cutoff-gradient term tends to zero in L^2, proving membership in $\mathcal{H}$ and extending the equation by density. The system gives $J_\phi = k|\nabla v|^2$. For $D > R_0$, the Laurent area formula and boundary integration give

$$\int_{D_D} k|\nabla v|^2 = \pi D^2 + o(1), \qquad \int_{D_D} v_Y = \pi D^2 - \pi \operatorname{Re} c.$$

Expand $|\nabla(v-Y)|^2+(k-1)|\nabla v|^2$ and let $D\to\infty$; this yields the last equality in (3.2). Alternatively, expanding $\Phi_k(w)=B_k+2\int b\cdot\nabla w+\|w\|_k^2$ and testing (3.1) with w yields its middle expression. Finally $b(\phi\circ h)=b(h)+c$, and admissibility bounds this real coefficient by the same supremum. This proves the asserted upper bound. □

For each finite K, the source's Cauchy–Beurling contraction works with $q=(k-1)/(k+1)$, not only an integer contrast: $g=q\overline{1+\mathcal{S}g}$ has contraction factor at most $(K-1)/(K+1)<1$. Its compact support yields a normalized exterior expansion. The passage from the weak solution to a principal quasiconformal *homeomorphism* uses the classical factorization and measurable Riemann mapping inputs used in the source. These are imported analytic inputs, not consequences of the energy minimization alone. Quasiconformal composition preserves the exact retained continuations and zero gradients using the chain rule and Lusin N property. The near-supremum hypothesis is genuinely global: maximizing over a finite collection of computed maps does not certify it.

Corollary 3.2 (Unequal finite contrasts)**.** *For disjoint measurable $F_j\subset F$ with $k\geq 1+\lambda_j$ almost everywhere on F_j, $\lambda_j>0$, the hypotheses above give*

$$\sum_j \lambda_j\int_{F_j}|\nabla v|^2\leq 2\pi\,\mathrm{Re}\,c-\int|\nabla(v-Y)|^2\leq 2\pi d. \tag{3.3}$$

The index set may be countable. In particular each zone has leakage at most $2\pi d/\lambda_j$.

Proof. Pointwise comparison with $(k-1)|\nabla v|^2$ proves the finite-sum claim. Monotone convergence proves the countable version. □

Remark 3.3. The corresponding variational residual arguments below also hold for symmetric matrix weights bounded below by the identity, with the appropriate matrix norms. We do not infer the scalar conjugacy system, coefficient identity, or admissible quasiconformal model class from that matrix generalization. Analytic and topological hypotheses must be checked separately.

4 Residual certificates in the physical energy norm

For $\psi\in\mathcal{H}$ let $\mathcal{R}_h(\psi)=-\int b\cdot\nabla\psi-a_k(w_h,\psi)$, where $w_h\in\mathcal{H}$ is any conforming candidate. Define

$$\|\mathcal{R}_h\|_{k,*}=\sup_{\|\psi\|_k=1}|\mathcal{R}_h(\psi)|.$$

There is no point-evaluation norm or sampled residual in this definition.

Theorem 4.1 (Residual and coefficient interval)**.** *Let w solve (3.1). For any $w_h\in\mathcal{H}$,*

$$\|w-w_h\|_k=\|\mathcal{R}_h\|_{k,*}\,,\qquad \Phi_k(w_h)=\Phi_k(w)+\|w-w_h\|_k^2\,. \tag{4.1}$$

Suppose $q\in L^2(\mathbb{R}^2;\mathbb{R}^2)$ is distributionally divergence-free. With $j=k\nabla w+b$ and

$$\eta^2=\int k^{-1}|q-k\nabla w_h-b|^2$$

one has the exact identity

$$\eta^2=\|w-w_h\|_k^2+\int k^{-1}|q-j|^2. \tag{4.2}$$

Under the additional map hypotheses of Proposition 3.1,

$$\boxed{\max\{0,\Phi_k(w_h)-\eta^2\}\leq 2\pi\,\mathrm{Re}\,c\leq\Phi_k(w_h).} \tag{4.3}$$

All constants are one in the displayed physical norms.

Proof. Subtract the equations to get $\mathcal{R}_h(\psi) = a_k(w - w_h, \psi)$. Cauchy–Schwarz gives the dual upper bound; the unit vector in the direction $w - w_h$ gives equality unless it is zero, when both sides vanish. Expand the quadratic functional Φ_k around w. Its linear term vanishes by (3.1); the remaining term is exactly $\|w_h - w\|_k^2$. The weak equation says that j is divergence-free. Write

$$q - k\nabla w_h - b = (q - j) + k\nabla(w - w_h).$$

The cross term in its k^{-1}-norm is $2\int (q - j) \cdot \nabla(w - w_h) = 0$. This follows first for compact smooth tests and then for $\mathcal{H}$ by density and the L^2 bounds. This proves (4.2). Subtract the error bound from (4.1), use nonnegativity of $\Phi_k(w)$, and apply (3.2) to obtain (4.3). □

This is a whole-plane specialization of the classical Prager–Synge orthogonality mechanism [5, Theorem 2.2], not a newly invented error principle. Its particular value here is that it certifies the source's normalized Laurent *gain* and leakage, rather than only an unspecified diffusion solution. Unlike the raw Cauchy–Beurling contraction estimate, its reliability constant does not grow as $(1 - \|q_{\text{Beltrami}}\|_\infty)^{-1}$. The weighted norm, the flux reconstruction cost, and matrix conditioning can still deteriorate with contrast. A constant-one theorem is not a contrast-independent complexity claim.

Inexact equilibrium and coefficients. For a flux not known to be equilibrated, set $\zeta = \sup_{\|\psi\|_k = 1} |\int q \cdot \nabla\psi|$. The same weak subtraction and Cauchy–Schwarz give

$$\|w - w_h\|_k \le \eta + \zeta. \tag{4.4}$$

Thus replace η^2 by $(\eta + \zeta)^2$ in the lower endpoint of (4.3). A finite collection of divergence moments is not an upper bound for ζ. If computation used $\widetilde{k}$ instead of k, the defect satisfies

$$\eta_k \le \left\| k^{-1/2}\{q - \widetilde{k}\nabla w_h - (\widetilde{k} - 1)e\} \right\|_2 + \left\| k^{-1/2}(k - \widetilde{k})(e + \nabla w_h) \right\|_2. \tag{4.5}$$

This follows by adding and subtracting the approximate constitutive field. It separates coefficient error from discretization and equilibrium error, including transmission interfaces. Exterior truncation needs its own continuation and tail bound; the integral in the theorem is over the entire plane.

Zone certification and inconsistency detection. Every certified upper bound $U \ge \Phi_k(w_h)$ gives $\sum_j \lambda_j \int_{F_j} |\nabla v|^2 \le U$ and $\int |\nabla(v - Y)|^2 \le U$. If a global near-maximality bound d is also independently available, use $\min\{U, 2\pi d\}$. A lower endpoint strictly exceeding $2\pi d$ proves incompatibility of the supplied certificates or hypotheses; it does not by itself identify which geometric or analytic premise failed.

4.1 Compatible Ram moments and unobserved residuals

Original TR218 explicitly fixes the operator domain, boundary coupling, and gauge before taking a matrix inverse. Original TR250's compatible variational development gives the physical Gram pullback and equilibrated flux identity; its proof and source-mismatch caveat were freshly read [6, 7]. They motivate the following implementation, whose validity is proved in physical space above.

Let $\psi_1, \dots, \psi_m$ have linearly independent gradients in $\mathcal{H}$. Define

$$A_{ij} = a_k(\psi_j, \psi_i), \quad f_i = -\int b \cdot \nabla\psi_i, \quad w_h = \sum_i c_i\psi_i, \quad \rho = f - Ac.$$

For an invertible moment matrix E on this finite span, $c = Rm$, $R = E^{-1}$, and the physical stiffness matrix is $R^T AR$, not the Euclidean identity by default. The restricted dual residual is

$$\sup_{0\neq\psi\in\operatorname{span}\{\psi_i\}} \frac{|\mathcal{R}_h(\psi)|^2}{\|\psi\|_k^2} = \rho^T A^{-1}\rho \le \|\mathcal{R}_h\|_{k,*}^2 . \tag{4.6}$$

Indeed write $\psi = \sum t_i\psi_i$ and maximize $(\rho^T t)^2/(t^T At)$; $t = A^{-1}\rho$ attains the maximum. This proves a *lower*, not upper, bound for the full residual. A Galerkin solve can make $\rho = 0$ while leaving a nonzero error orthogonal to the entire finite span.

If a reconstructed flux is a global rotated gradient $q = (\partial_Y p, -\partial_X p)$ with p in the homogeneous Sobolev space, then $\operatorname{div} q = 0$ by commutation of distributions. Piecewise polynomial fluxes instead require matching normal traces on every interface and an appropriate exterior continuation. Polynomial moments can compute their weighted defect norm exactly on rational panels when the coefficient and geometry permit exact integration. Moment reconstruction, full energy integration, and exact equilibrium are three different checks.

The volume21 convergence proof [8] distinguishes strong testing on each fixed vector from norm convergence over all unit vectors. That distinction applies here: refining a finite bank does not, without density and uniform control, certify an infinite residual or the all-end compactness statement. A large multiple of a vector in the orthogonal complement of a finite bank has all its probes zero. This observation is a limitation, not an additional contribution.

4.2 An exact disk fixture

Take $F = \overline{D}_a$, $k = \kappa \ge 1$ on F, and $k = 1$ outside. Set $\theta = (\kappa-1)/(\kappa+1)$. The exact solution is

$$\phi(z) = \begin{cases} (1+\theta)X + i(1-\theta)Y, & |z| \le a, \\ z + \theta a^2/z, & |z| \ge a. \end{cases}$$

The two formulas agree on the circle. The interior is an invertible affine map for finite κ; the exterior map is univalent because its difference at z_1, z_2 is $(z_1 - z_2)(1 - \theta a^2/(z_1 z_2))$. Its image is the exterior of the matching ellipse, proving the global homeomorphism property. Direct differentiation verifies the conductivity system and normal-flux matching. Polar integration gives

$$c = \theta a^2, \qquad \int |\nabla(v - Y)|^2 = 2\pi a^2\theta^2, \tag{4.7}$$
$$(\kappa-1)\int_F |\nabla v|^2 = \frac{4\pi a^2(\kappa-1)}{(\kappa+1)^2}, \qquad \|w\|_k^2 = \pi a^2(\kappa+1)\theta^2.$$

For example, the exterior error is the dipole $w = -\theta a^2 Y/|z|^2$, whose squared-gradient integral outside D_a is $\pi a^2\theta^2$; the interior contributes the same. The sum in (3.2) is $2\pi a^2\theta$, as required. For $a = 1, \kappa = 3$ and $w_h = w/2$, the exact equilibrated flux $q = j$ gives $\eta^2 = \pi/4$, $\Phi_k(w_h) = 5\pi/4$, and the coefficient interval $[\pi, 5\pi/4]$ for $2\pi \operatorname{Re} c = \pi$. This is a manufactured check of a certificate, not a method for discovering an unknown flux. The ellipsoidal image is admissible for the source's Jordan-disk model class, but is not generally a circle-domain target.

Proposition 4.2 (Conditional heterogeneous high-contrast limit)**.** *Fix a single compact omitted set F and a single source model h. Suppose finite contrasts k_n give the homeomorphisms of Proposition 3.1, with all $\phi_n \circ h$ admissible and $\Phi_{k_n}(w_n) \le U < \infty$ independent of n. Suppose measurable sets F_j cover F up to a planar null set and $\operatorname{ess\,inf}_{F_j}(k_n - 1) \to \infty$ for every fixed j. Then a subsequence of v_n converges locally uniformly and weakly in $W^{1,2}_{\mathrm{loc}}$ to a continuous v with*

$\int |\nabla(v-Y)|^2 \le U$ *and* $\nabla v = 0$ *almost everywhere on* F*. If the nonpoint components of* F *are closed Jordan disks,* v *is constant on every component.*

Proof. The energy bound gives uniform local gradient bounds for v_n. Coordinates of homeomorphisms attain extrema on the boundary of every relatively compact open set. For almost every centered circle, the one-dimensional Sobolev inequality and then integration in $\mathrm{d}t/t$ give

$$(\operatorname{osc}_{\overline{D}(x,r)} v_n)^2 \log(R/r) \le 2\pi \int_{D(x,R)} |\nabla v_n|^2 \quad (0 < r < R).$$

Exterior univalence and principal normalization on the common exterior of F give $|v_n - Y| \le C/|z|$, uniformly for large $|z|$, by the area theorem. Boundary extrema then bound v_n on each compact set. The displayed modulus and compactness give local uniform convergence; the gradient bounds give a compatible weak limit. Lower semicontinuity gives the error bound. For fixed j the leakage bound tends to zero; weak convergence on a disk containing F implies the limiting gradient vanishes on F_j. Take the countable union and its null complement. A Sobolev function with zero gradient on a connected open disk interior is constant there, and continuity extends that value to its boundary. Points require no argument. □

If the F_j cover only the nonpoint disk interiors, the conclusion about all of F does not follow: omitted point dust may have positive area. If F itself varies, this proposition requires additional set convergence and identification hypotheses. No uniform quasiconformal distortion bound in n was assumed or obtained. None of these certificates constructs a globally nearly maximizing model.

5 Weighted filling and two-sided barrier payments

5.1 A midpoint improvement in finite hole filling

Lemma 5.1 (Weighted jump ledger)**.** *Let* v *be bounded and locally Sobolev with finite Dirichlet energy on a finite circle domain containing infinity. At each disk hole* T_j *of radius* r_j*, assume its entire cluster range lies in a known interval of length* A_j*. Fill that hole by the interval midpoint, obtaining* $\widetilde{v}$*, and extend the domain gradient by zero to* g*. For* $\varphi \in C_c^\infty(\mathbb{C})$ *and* $\ell = 1, 2$*,*

$$\left| \int \widetilde{v}\, \partial_\ell \varphi + \int g_\ell \varphi \right| \le 2 \|\varphi\|_\infty \sum_j r_j A_j \le 2 \|\varphi\|_\infty \left(\sum_j \frac{r_j^2}{\omega_j} \right)^{1/2} \left(\sum_j \omega_j A_j^2 \right)^{1/2} \tag{5.1}$$

for every choice $\omega_j > 0$*.*

Proof. On almost every axial line the restriction is absolutely continuous on domain intervals and has integrable derivative on bounded portions. At each of the two endpoints of a disk interval its jump to the midpoint is at most $A_j/2$. Its transverse projection has length $2r_j$, so its integrated contribution is at most $2r_j A_j$. There are only finitely many point holes, affecting a null set of transverse parameters. Integration by parts on the intervals, followed by Fubini, proves the first bound. Apply Cauchy–Schwarz to $(r_j/\sqrt{\omega_j})$ and $(\sqrt{\omega_j} A_j)$ for the second. □

The source's arbitrary choice within a cluster interval gives the valid constant 4 instead of 2. Midpoint filling halves this particular distributional error without changing retained core values. It does not make the filled function continuous or Sobolev at a fixed stage. On an exhaustion, a bounded weighted radius sum and a vanishing weighted oscillation sum make this jump term vanish. The source's contour argument is still needed to identify the weak limit on positive-area omitted dust. Point samples of a boundary function do not certify its *entire* cluster interval.

5.2 Two paths from the plateau to zero

Theorem 5.2 (Weighted slice resistance)**.** *Let V be continuous and locally Sobolev on a neighborhood of a rectangle $[-L,L]\times I$, $L>0$. For almost every y in a measurable $J\subset I$, suppose some $x_y\in(-L,L)$ satisfies*

$$V(x_y,y)\ge 1-\epsilon(y),\qquad |V(-L,y)|,|V(L,y)|\le\delta(y),\qquad \epsilon,\delta\ge 0.$$

Put $g(y)=(1-\epsilon(y)-\delta(y))_+$. Let $a(x,y)>0$ be measurable, with finite positive total resistance $\mathscr{R}(y)=\int_{-L}^{L}a(x,y)^{-1}\,\mathrm{d}x$ for almost every $y\in J$. Then

$$\int_J\int_{-L}^{L} a|\partial_X V|^2\,\mathrm{d}x\,\mathrm{d}y\ge\int_J\frac{4g(y)^2}{\mathscr{R}(y)}\,\mathrm{d}y. \tag{5.2}$$

If the point locations are specified and the two resistances are $\mathscr{R}_-,\mathscr{R}_+$, a sharper integrand is $g^2(\mathscr{R}_-^{-1}+\mathscr{R}_+^{-1})$.

Proof. Use a good ACL line; continuity identifies the Sobolev representative with the values in the hypotheses. Weighted Cauchy–Schwarz on each side of x_y gives

$$\int_{-L}^{x_y} a|V_X|^2\ge g^2/\mathscr{R}_-,\qquad \int_{x_y}^{L} a|V_X|^2\ge g^2/\mathscr{R}_+.$$

The case of zero g is immediate. Otherwise both resistances are positive, and $\mathscr{R}_-^{-1}+\mathscr{R}_+^{-1}\ge 4/(\mathscr{R}_-+\mathscr{R}_+)$. This bound is independent of which available x_y is chosen, so no measurable selection is needed for (5.2). Integrate over J. The same proof applies when the energy is infinite. □

For $a=1$, this is

$$\int|\nabla V|^2\ge\frac{2}{L}\int_J g(y)^2\,\mathrm{d}y. \tag{5.3}$$

With an exact plateau and zero endpoint traces it improves the source bound $|J|/(2L)$ to $2|J|/L$, by using *both* paths to zero. The constant is sharp for the one-dimensional information: the tent function with its peak at the midpoint has energy $2/L$. For unequal weights equality in the one-dimensional Cauchy inequalities is attained by functions affine in resistance coordinates. For a disk support D_L one can use chord endpoints $\pm\sqrt{L^2-y^2}$ and obtain the still sharper bound

$$\int|\nabla V|^2\ge\int_J\frac{2g(y)^2}{\sqrt{L^2-y^2}}\,\mathrm{d}y.$$

This is a directional energy certificate, not a claim that a two-dimensional barrier attains all linewise optima simultaneously.

For a positive-definite matrix weight A, the pointwise inequality $|e_1\cdot\nabla V|^2\le(e_1^TA^{-1}e_1)(\nabla V^TA\nabla V)$ proves the same lower bound for $\int\nabla V^TA\nabla V$, with $\mathscr{R}(y)=\int e_1^TA^{-1}e_1\,\mathrm{d}x$. The matrix must describe the physical energy; an arbitrary coefficient-space norm is not such a weight.

Corollary 5.3 (Finite geometric exclusion)**.** *Let a continuous global Sobolev barrier be supported in D_L, and equal one on a closed contour surrounding a compact continuum K. If its energy is at most E, each axial projection interval of K has length at most $LE/2$, and*

$$\operatorname{diam}K\le LE/\sqrt{2}.$$

For a plateau lower bound $1-\epsilon$ and outer traces of magnitude at most δ, replace E by $E/(1-\epsilon-\delta)^2$ when $\epsilon+\delta<1$.

Proof. Every transverse line through a projection of K meets the surrounding contour. Apply (5.3) in each direction. The diameter is at most the square root of the sum of squared projection widths. □

In the source's contradiction argument, a barrier may depend on the chosen component and stage. The corollary then bounds that component; it does not provide one finite list covering all possible limiting components. Vanishing energies on saved contours at arbitrarily late stages are still required. Finite positive errors give exclusion above a stated scale, not exact singleton components.

6 Sharper finite-tail flux and Laurent stability

Assume an actual conformal equivalence $f : \Omega \to \Omega'$ of circle domains, with infinity in both, normalized by

$$f(z) = z + \sum_{n\geq 1} \alpha_n z^{-n}.$$

Index every component pair for which at least one side has positive radius as $(K_j, K_j') = (\overline{D}(a_j, r_j), \overline{D}(b_j, s_j))$. Zero radii are allowed. Assume the source's upper-semicontinuous all-component correspondence and its null common-point trace projections on almost every axial line. These are the exact geometric inputs to the slice argument; we do not infer them from numerical samples or prove their global removable-boundary antecedent here. Choose R strictly containing the full source complement, and put

$$A = \pi R^2, \quad B_R = \int_{\Omega\cap D_R} |f'|, \quad \tau_N = \sum_{j>N} r_j s_j, \quad T_R = \pi \sum_{n\geq 1} n|\alpha_n|^2 R^{-2n}.$$

Disjoint disk interiors and bounded complements give square summability of both radius sequences and hence $\tau_N < \infty$.

Theorem 6.1 (Shrinking enlargements with summable domination)**.** *Under these hypotheses, for every finite initial list of size $N \geq 0$,*

$$2\pi R^2 \leq 2B_R + 2\pi \sum_{j\leq N} r_j s_j + 8\tau_N. \tag{6.1}$$

Thus the source's finite-tail constant 16 *can be replaced by* 8*. No positive separation uniform over the tail is assumed.*

Proof. Remove slightly enlarged concentric disks about the first N source components, allowing positive enlargement of source point components. They can be mutually disjoint and contained in D_R. As in the source, the enlargements can be chosen so that every remaining positive-radius component meets at most one enlarged disk: retain a positive gap between the finite initial components; sufficiently small remaining disks cannot bridge that gap; the finitely many larger remaining disks can be avoided individually. Shrinking further preserves this property. Choose a decreasing sequence of such enlargements with radii tending to the original radii.

On a component segment of a retained axial slice, the closed-graph component relation has interval projection fibers. Their union covers the interval between the endpoint values: otherwise a missing value would separate the segment into two open classes. Domain arcs pay their integral of $|f'|$, common-point traces pay zero by the stated nullity hypothesis, and a met tail component pays at most $2s_j$. This recovers the source's slice estimate without a pointwise boundary extension assumption.

For each direction let $c_{j,m}$ be the integral over transverse parameters of $2s_j$ times the number of retained slice segments meeting K_j at enlargement stage m. A line intersects a disk in an interval. Removing at most one intersecting enlarged disk leaves at most two pieces, so

$$0 \le c_{j,m} \le 8r_j s_j.$$

For a *fixed* tail disk K_j, its distance from each of the first N compact components is positive. It therefore meets none of their enlargements once m is sufficiently large, depending on j. Thereafter its interval lies in at most one retained segment and $c_{j,m} \le 4r_j s_j$. Source point components have zero integrated cost: their exceptional lines are countable.

Summable domination now gives

$$\limsup_{m\to\infty} \sum_{j>N} c_{j,m} \le 4\tau_N \tag{6.2}$$

in each direction. Explicitly, split the series at a finite index M, use eventual multiplicity one for its finite head, and bound the remainder by $8\sum_{j>M} r_j s_j$; then let $M \to \infty$. This argument is the step absent from the source's coarser constant. It does not interchange a limit with an undominated infinite sum.

Integrating both slice directions gives the outer flux minus the inner fluxes and a domain contribution at most $2B_R$. The outer flux is exactly $2\pi R^2$ by the Laurent normalization. On the jth enlarged circle, subtracting the target center and using component correspondence gives inner flux limsup at most $2\pi r_j s_j$, including either zero-radius case. There are only N such inner terms. Use (6.2) twice while the enlargements shrink. This proves (6.1). For $N = 0$ no enlargement is needed and every multiplicity is already one. Finite and empty pair lists are covered by empty tails. □

Theorem 6.2 (Finite-budget defect certificate)**.** *Define the nonnegative source and target area slacks*

$$D_s = A - \int_{\Omega\cap D_R} 1 - \pi \sum_j r_j^2,$$
$$D_t = A - T_R - \int_{\Omega\cap D_R} |f'|^2 - \pi \sum_j s_j^2.$$

Under the preceding hypotheses,

$$\boxed{\int_{\Omega\cap D_R} (|f'| - 1)^2 + \pi \sum_j (r_j - s_j)^2 + T_R + D_s + D_t \le (8 - 2\pi)\tau_N.} \tag{6.3}$$

More generally, if the flux inequality is certified only with an additive error $\varepsilon_f \ge 0$, and the two area upper bounds have additive errors $\varepsilon_s, \varepsilon_t \ge 0$, then

$$\int_{\Omega\cap D_R} (|f'| - 1)^2 + \pi \sum_j (r_j - s_j)^2 + T_R \le \Gamma_N := (8 - 2\pi)\tau_N + \varepsilon_f + \varepsilon_s + \varepsilon_t. \tag{6.4}$$

The slacks in (6.3) are asserted only for the exact areas.

Proof. The source pieces are disjoint subsets of D_R, proving $D_s \ge 0$. The outer image contour encloses the actual image of $\Omega \cap D_R$ and every target complementary component. Its Laurent area

is $A - T_R$. Conformal change of variables and disjointness of disk interiors prove $D_t \geq 0$, with no area-zero assumption on target point dust. Set $J_R = B_R + \pi \sum_j r_j s_j$ and expand:

$$\int_{\Omega \cap D_R} (|f'| - 1)^2 + \pi \sum_j (r_j - s_j)^2 + T_R + D_s + D_t = 2A - 2J_R. \tag{6.5}$$

Subtract $2J_R$ from (6.1); the remaining coefficient is $8 - 2\pi > 0$. This proves (6.3). If instead the two area inequalities have the stated errors, their sum bounds the same expanded expression without the slacks by $2A + \varepsilon_s + \varepsilon_t - 2J_R$. The approximate flux bound then proves (6.4). □

Certified squared-radius tails $R_N^2 \geq \sum_{j>N} r_j^2$ and $S_N^2 \geq \sum_{j>N} s_j^2$ give $\tau_N \leq R_N S_N$. This is useful only if these are bounds on the *complete* tail. Likewise, ε_f must include any unaccounted common-point trace or slice defect; zero sampled trace errors do not set it to zero. With the exact source hypotheses and $N \to \infty$, the defect vanishes and the source's equality argument applies. The improvement here is the finite budget and the smaller tail constant, not a replacement of the trace-null theorem.

Corollary 6.3 (Exterior map and derivative control)**.** *If $T_R \leq \Gamma$ is certified and $|z| > R$, then*

$$\begin{aligned} |f(z) - z| &\leq \sqrt{\Gamma/\pi}\, \sqrt{-\log(1 - R^2/|z|^2)}, \\ |f'(z) - 1| &\leq \sqrt{\Gamma/\pi}\, \frac{R}{|z|^2 - R^2}. \end{aligned} \tag{6.6}$$

For a rationally convenient weaker value bound, replace the logarithm by $R^2/(|z|^2 - R^2)$.

Proof. Apply Cauchy–Schwarz to the Laurent series, with energy weights nR^{-2n}. The complementary value series is $\sum_{n\geq 1} (R/|z|)^{2n}/n = -\log(1 - R^2/|z|^2)$. For the derivative it is $\sum_{n\geq 1} nR^{2n}|z|^{-2n-2} = R^2/(|z|^2 - R^2)^2$. Absolute convergence justifies termwise differentiation outside the circle. Finally $\sum t^n/n \leq \sum t^n = t/(1-t)$ for $0 < t < 1$. □

The exterior conclusion controls the complex derivative, not merely its modulus. It uses the entire Laurent norm and the normalization at infinity. It does not yield uniform control on an arbitrary rough boundary or across thin interior corridors; its constants diverge as $|z| \downarrow R$. Rotations and translations must be removed by the stated normalization before interpreting closeness to the identity.

7 Worked finite certificates and their limits

7.1 A conforming polynomial panel with a hidden mode

Let $Q = (-1, 1)^2$, let $k = \kappa > 1$ on Q and $k = 1$ outside, and extend the polynomial $\psi = (1 - X^2)(1 - Y^2)$ by zero outside Q. It is globally H^1 with compact support. For rational parameters t, s, take $w_h = t\psi$ and $q = s(\partial_Y \psi, -\partial_X \psi)$. The flux is globally divergence-free, not merely elementwise so. Exact monomial integration gives

$$G = \int |\nabla \psi|^2 = \frac{256}{45}, \qquad \int_Q e \cdot \nabla \psi = \int_Q e \cdot (\partial_Y \psi, -\partial_X \psi) = 0, \tag{7.1}$$

$$\Phi_k(w_h) = 4(\kappa - 1) + \kappa t^2 G, \qquad \eta^2 = \frac{s^2 G + \kappa^2 t^2 G + 4(\kappa - 1)^2}{\kappa}. \tag{7.2}$$

For example, $\int_{-1}^{1} X^2\,\mathrm{d}X = 2/3$ and $\int_{-1}^{1}(1-Y^2)^2\,\mathrm{d}Y = 16/15$, giving $128/45$ for each derivative component. The cross term between a gradient and its rotated gradient vanishes pointwise; the other cross terms vanish by oddness. This proves both formulas.

With $s = t = 0$, the true coefficient gain of the principal conductivity map is consequently bracketed by

$$\frac{4(\kappa-1)}{\kappa} \le 2\pi \operatorname{Re} c \le 4(\kappa-1). \tag{7.3}$$

In contrast, the Galerkin residual against the entire one-dimensional space $\operatorname{span}\{\psi\}$ is exactly zero at $w_h = 0$. The continuum right-hand side is not zero: a compact smooth test equal to Y on a neighborhood of $\overline{Q}$ has $-\int_Q (\kappa-1)e \cdot \nabla\psi_{\text{test}} = -4(\kappa-1)$. Hence the true gradient is nonzero. This is an actual domain-level example of a missing mode, not only an abstract matrix nullspace.

As a concrete Ram moment, use $m = \int_Q w_h$. Since $\int_Q \psi = 16/9$, the synthesis is $t = 9m/16$ and the physical stiffness becomes

$$\kappa G(9/16)^2 = \frac{9\kappa}{5}.$$

This weighted moment is continuous and reconstructs the declared one-dimensional span exactly. It supplies no guarantee on its infinite-dimensional complement. For $\kappa = 3$, (7.3) is $8/3 \le 2\pi \operatorname{Re} c \le 8$. At very high contrast the upper bound is loose; known exterior area bounds can be intersected with it, but finite moment invertibility itself does not sharpen the continuum bound. The square fixture does not assert that this deformation has a circle-domain image or that it is nearly extremal.

7.2 No uniform tail separation is needed

Place a first source disk at $\overline{D}(0,1)$, and for $j \ge 1$ place a tail disk of radius $r_j = 2^{-j}/4$ at $1 + 2^{-j}$ on the real axis. The tail disks are mutually disjoint, and their distances from the first disk are $3 \cdot 2^{-j}/4$. Enlarge the first disk by 2^{-m}. Exactly the tail indices $j \ge m$ meet this enlargement (tangency conventions do not affect the argument). For illustrative target-radius weights $s_j = 2^{-j}/3$, the one-direction domination used in the proof is

$$4\sum_{j\ge 1} r_j s_j + 4\sum_{j\ge m} r_j s_j = \frac{1}{9} + \frac{4^{1-m}}{9} \longrightarrow \frac{1}{9}.$$

No enlargement avoids the entire tail. This explicit summable model explains why uniform separation is neither available nor needed. The s_j here are only weights illustrating the summation estimate; no conformal correspondence with these prescribed target radii is claimed.

7.3 Exterior control without an invalid rigidity example

For $f(z) = z + \frac{1}{4z}$ on $|z| > 1$, univalence follows from the difference factorization used in Section 4.2. At $R = 1$, $T_R/\pi = 1/16$. At $z = 2$ the exact derivative error is $1/16$ and the bound in (6.6) is $1/12$. The exact value error is $1/8$, whose square $1/64$ is below the rational upper bound $1/48$. The target boundary is an ellipse, not a circle. Thus this example checks only the exterior coefficient estimates; it is not a nontrivial example satisfying all the circle-domain flux hypotheses.

8 Comparison, safeguards, and future work

Source step	Local improvement	Retained obligation
Constant n on the full omitted set	Variable contrast, zone leakage, and coefficient interval	Actual principal homeomorphism, exact retained tests, and global near-maximality when d is used
Hole filling with any cluster value	Midpoint constant 2 and weighted oscillation ledger	Full cluster intervals, complete hole list, and contour value identification in the limit
One-path payment $\lvert J\rvert/(2L)$	Two-path cost $2\lvert J\rvert/L$, weighted resistance, and trace-error tolerance	Continuous Sobolev representative, all relevant lines, and actual surrounding contour
Finite flux tail $16\tau_N$	$8\tau_N$ by fixed-component convergence and summable domination	All-component correspondence and null common-point traces
Nonnegative equality after discarding Laurent defect	Budget for radii, bulk modulus, area slacks, and exterior map error	Exact normalization, actual-image area bounds, complete radius tails

Corrections versus refinements. No counterexample to a displayed local source identity was established in this scoped study. The constants 4 in hole filling, $1/(2L)$ in the one-path barrier bound, and 16 in the source's finite-tail flux bound are valid but weaker; the changes above are refinements, not repairs of false inequalities. The global source claim is not independently certified, because its companion construction is imported and its status conflicts with the contemporary survey's open-problem formulation. A full proof audit would have to resolve that dependency and examine every external regularity and compactness input.

What is gained. The energy functional supplies a coefficient interval with a correct whole-plane gauge and no unknown reliability constant. Heterogeneous contrasts expose where leakage is actually penalized. The barrier certificate converts an energy tolerance into an excluded projection scale, with explicit weight and trace dependence. Most significantly for the source's final argument, the tail summation improvement and retained Laurent defect give a smaller finite budget and genuine complex-analytic exterior control. These are source-specific changes to quantitative conclusions, not credit for renaming a Gram matrix.

Costs and failure modes. Constructing an equilibrated global flux can be harder than solving a finite primal system. High contrast still affects conditioning and the sharpness of a computed estimate. A nearly maximizing model is not supplied by residual minimization. Energy smallness in dimension two gives no general uniform norm bound without additional structure; the source's homeomorphism-coordinate modulus is essential in its height limit. A finite barrier only rules out sufficiently large components. A small nonzero flux deficit never proves exact Möbius rigidity. The exterior estimates do not control all boundary values, and an all-components tail cannot be replaced by retained large disks plus an unverified drawing of the dust.

Extensions that are feasible but not completed here. One can combine conforming polynomial panels with stream-function fluxes and interval integration to construct a concrete certified solver for a specified finite conductivity profile, including an analytic exterior continuation. An adaptive scheme should separately budget constitutive error, equilibrium error, exterior truncation, cluster-interval uncertainty, and the full radius tail. Weighted resistance estimates can exploit anisotropic physical energy bounds when these are actually available. Improving 8 further would

require more geometric information than the present full projection interval payment; no optimality of that planar constant is asserted. The major mathematical task remains validating or replacing the companion retained-test selection with all its uniform and actual-image properties. None of the finite certificates removes that task.

Contribution calibration. The C3 assessment concerns the combined variable-contrast coefficient certificate, two-sided weighted exclusion, and the sharper finite-tail Laurent stability theorem. Their tools are classical coercivity, orthogonality, Cauchy–Schwarz, and dominated convergence. The assessment does not assert historical novelty, an unconditional proof of the remaining rigidity conjecture, or a new theorem obtained merely by changing to Ram coordinates. Applicability L3 is an independent catalogue classification.

9 Reading and verification boundary

All 10 cached TeX build files, the bibliography, and the source README were literally read in bounded displays. The main scientific chains in all seven section modules were read completely, including the normalization, finite-transfer interface, point-dust values, coarea representatives, saved-barrier limits, masks, and flux multiplicity. This is a manuscript read, not a read of its companion or the full literature it cites.

Fresh private reading comprised original TR218 lines109–491; the complete volume21/chapter3 convergence component; TR250's foundation revision lines121–275 and465–644; and the comparative TR-item-343 passages181–267,492–686,852–905. These are the actual intervals used; unread complements remain explicit in the separate hashed read ledger. TR218 supplies real operator and boundary derivations, and TR250 supplies actual variational and residual proofs, rather than only a comparative index. TR-item-343 contributes the distinction between actual-image identification and all-component compactness, not a substitute retained-test construction. No whole-corpus read is asserted.

The independently executable exact checks accompanying this report use rational arithmetic for disk identities after cancellation of π, polynomial panel moments, finite weighted orthogonality and residual identities, resistance inequalities, shrinking-tail fixtures, defect algebra, and rational exterior bounds. Their purpose is to catch sign, factor, and normalization errors in the proved formulas. They do not test removability, all common-point traces, uncountably many components, or the existence of a global conformal map. Counts, units, exact scripts, pre/post endpoint hashes, native and fallback compile records, and all-page visual inspection are recorded separately in the owned audit files. Independent cross-review checks are not counted as author tests.

References

[1] OpenAI, *Removable Boundaries and Rigidity of Circle Domains*, September 23, 2026. Pinned public build at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Removable-Boundaries-and-Rigidity-of-Circle-Domains-September-23-2026/paper.pdf.

[2] OpenAI, *Koebe's Circle-Domain Conjecture*, September 23, 2026. Theorem7.5 and Corollary7.6 are imported through the preceding source; the companion proof was not independently read for this report. https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e

4b4c146bf78a/preprints/Koebes-Circle-Domain-Conjecture-September-23-2026/paper.pdf.

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[5] F. Bertrand and D. Boffi, *The Prager–Synge theorem in reconstruction based a posteriori error estimation*, arXiv:1907.00440v1, 2019, Section2, especially Theorem2.2 and its proof. https://arxiv.org/html/1907.00440v1.

[6] M. Subbarao, *Green Functions, Resolvents, and Boundary Traces in Ram Calculus*, original private TR218. Locally read proof scope lines109–491; no public upload.

[7] M. Subbarao, *Compatible Variational Ram Panels*, private foundation revision `TR-rev-1-old-tr250.tex`. Actual proof scope lines121–275 and465–644.

[8] M. Subbarao, *RamCalBooks*, volume21, chapter3, `convergence.tex`, complete 183-line component locally read. Other volume components are not claimed read.

[9] M. Subbarao, private comparative report `TR-item-343-astra.tex`, scoped passages on actual-image and all-component compactness, with the exact local ranges above. No proof of the retained-test selection is imported from this report.